

Adivi Arun

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EDUCATION

Indian Institute of Science Bangalore

Nov. 2021 – May 2026

Integrated BS-MS (Research) - Biology Major

CGPA : 8.5/10.0

Oxford Independent PU College Bangalore

May. 2019 – Oct. 2021

Karnataka Pre-University Certificate Examination : 596/600

NPS Rajajinagar Bangalore

May. 2007 – May. 2019

Institute Topper

All India Secondary School Examination: 495/500

RESEARCH EXPERIENCE

Predicting Translational Readthrough from stop-codon sequence context

July 2025 - May 2026

*Eswarappa Lab - IISc | Advisor : Prof. Sandeep Eswarappa and Prof. Nagasuma Chandra | **Masters Thesis***

- Used the mRNA-LM transformer language model on mRNA sequence data and trained a linear classifier on the embeddings to generate readthrough predictions.
- Investigated the effect of varying nucleotide context window sizes on model performance and demonstrated an advantage in using the entire 3'UTR and CDS sequences for predictive accuracy.
- Used the prediction head of the mRNA-LM model to directly generate predictions and applied LoRa-based finetuning to adapt the model.

Predicting Visual Behavior in Stroke Patients using Topographic Neural Networks

July - Dec. 2024

*NeuroAI Lab - EPFL | Advisor : Dr. Johannes Mehrer | **Bachelors Thesis***

- Created perturbation modules in a topographic neural network model to simulate the effects of lesions in specific brain locations post-stroke.
- Tested the model on benchmarks of human visual perception, like the HVF test, and compared its performance to stroke subjects from human studies.
- Summarised all results, methodologies, and analyses into a comprehensive graduate-level Thesis report

Effect of reward landscape on the rate of motor learning in continuous action spaces

July - August 2022

Dhawale Lab - Centre for Neuroscience | Advisor: Prof. Ashesh Dhawale

- Simulated motor learning on steep and shallow gradients in a continuous action space.
- Implemented value-based methods like Q-learning and Thompson Sampling, and Policy Gradient learning algorithms in MATLAB.
- Fitted exponential functions to the learning curves and derived time-constants.
- First-author research selected for a Poster Presentation in the **Women in Machine Learning Workshop** as part of the **NeurIPS'23 Conference**.

Recurrent Neural Networks for Meta-Reinforcement Learning

May - June 2023

Dhawale Lab - Centre for Neuroscience | Advisor: Prof. Ashesh Dhawale

- Investigated an artificial RL agent's ability to infer task structure and adapt to novel environments while learning, similar to the brain.
- Built and trained Google DeepMind's LSTM network architecture and the Advantage Actor-Critic learning algorithm in PyTorch.
- Simulated dependent and independent multi-armed bandit environments and investigated for meta-learning in the dynamics of the RNN.

Psychophysics of Visual Search

May - June 2024

Kreiman Lab - Harvard University | Advisor : Prof. Gabriel Kreiman

- Conducted a literature review on visual search behaviour and the different stimulus attributes that guide it.
- Worked on testing the effects of target distractor similarity on a computational visual search model and quantifying alignment with human behaviour.

RELEVANT COURSEWORK

University Courses:

- **Bioinformatics:** Evolutionary Genetics, Bioinformatics, Principles of Genetic Engineering, Computational Epidemiology, Stem Cell Technology, Cell Mechanics
- **Computer Science:** Algorithms and Programming, Optimisation Methods, Scientific Computing (audit)
- **ML and AI:** Machine Learning for Data Science, Introduction to NLP
- **Mathematics:** Analysis and Linear Algebra I and II, Probability and Statistics

ACADEMIC ACHIEVEMENTS

EPFL SRP Intern: Selected for the Summer Research Program at the School of Life Sciences at EPFL. One among **24** accepted candidates among over 800 applicants worldwide.

Khorana Scholar: Among a group of Indian students selected for research internships in the field of Biotechnology and Computational Biology at prestigious US universities as part of a program by the Govt of India and the Indo-US Science and Technology Forum.

KVPY Scholar: Ranked **71** among over 100,000 candidates. Awarded the KVPY scholarship by the Department of Science and Technology, Govt of India

National Talent Search Examination Scholar: Awarded a Scholarship by the Government of India for placing in the **top 1000** among over 1.4 million students in the country.

Regional Mathematics Olympiad: Qualified for the Indian National Mathematics Olympiad and was one among **30** candidates invited to attend a training camp by the Indian Statistical Institute.

Junior Science Olympiad: Awarded a Certificate of Merit for placing among the **top 1%** of students in the country.

Indian Computing Olympiad: Qualified at the Zonal Level and placed among the **top 100** candidates nationwide.

Indian Linguistics Olympiad: Qualified at the national level and was among **35** candidates selected for a training program at IIIT Hyderabad.

SKILLS

Programming Languages

- Python - NumPy, Pandas, PyTorch, Scikit-learn, Brainscore libraries
- MATLAB - Psychtoolbox and Signal Processing Toolbox
- R and R Studio
- C and C++
- LaTeX

Laboratory Skills: Gel Electrophoresis, Spectrophotometry, Cell culture plating, Light Microscopy, PCR

CO-CURRICULAR ACTIVITIES

- Women in Machine Learning (WiML) Workshop volunteer
- NeurIPS'23 Conference volunteer
- Event Coordinator of Dianoia and ACC for Pravega - a nationwide cultural and sci-tech fest
- Member of the college athletics, throw-ball, and handball teams.
- Trained in Indian classical music for over 5 years and gave several stage performances